

What is Power BI?

Power BI is a business intelligence (BI) tool by Microsoft that helps organizations to connect, visualize and analyze data. It allows users to create interactive dashboards and reports, so they can make data-driven decisions easily.

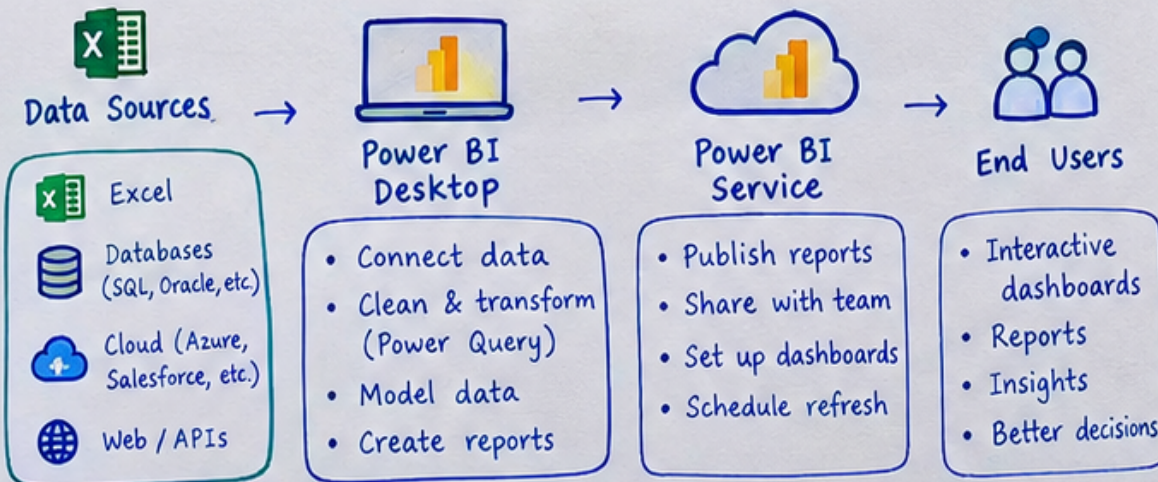
- It can connect to multiple data sources (Excel, databases, cloud, etc.)
- It transforms raw data into meaningful insights using visuals and reports.
- It is a self-service BI tool, meaning even non-technical users can use it.



Power BI
by Microsoft

"From data to insights"

How Power BI Works?

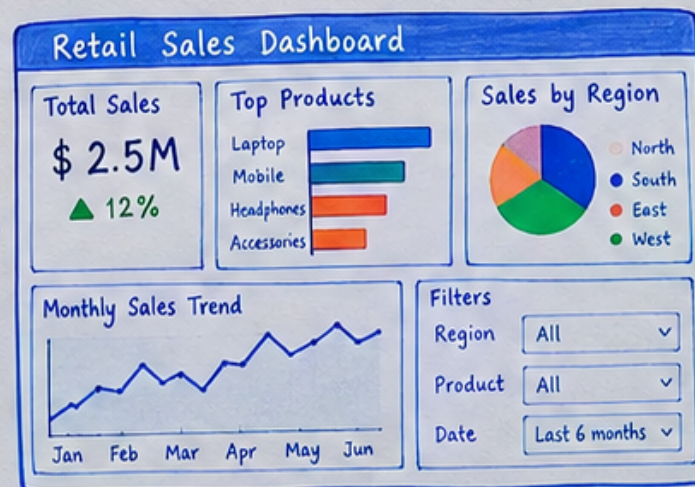


Example: Retail Company

A retail company wants to understand its sales performance across different regions, products and time periods.

Using Power BI, they can:

- Connect data from multiple sources (Sales database, Excel, CRM).
- Create interactive dashboards showing total sales, top products, region-wise performance, etc.
- Allow managers to filter by date, region or product category.
- Get real-time insights to take quick decisions (e.g., increase stock for high-demand products, run offers in low-performing regions).



Why Companies Use Power BI Instead of Other BI Tools?

Many companies have options like Tableau, Qlik, Looker, etc. But Power BI is widely chosen because it offers a unique combination of features, value and ease of use. Here are the key reasons:

- ① **Cost-Effective** – Power BI has lower licensing costs compared to many other enterprise BI tools.
- ② **Easy to Use** – Simple and intuitive interface, which reduces the learning curve.
- ③ **Seamless Integration** – Works smoothly with Microsoft products (Excel, Azure, Teams, SharePoint, etc.).
- ④ **Powerful Data Modeling** – Strong DAX language and data transformation capabilities.
- ⑤ **Scalable** – Handles small to large amounts of data and grows with the business.
- ⑥ **Rich Visualizations** – Offers a wide range of interactive and customizable charts and dashboards.
- ⑦ **Real-time Insights** – Supports real-time data and live dashboards.
- ⑧ **Strong Community & Support** – Large user community and continuous updates from Microsoft.

In Short...

Power BI is not just a tool, it's a complete ecosystem that helps businesses turn their data into valuable insights. Companies prefer it because it is affordable, easy to use, integrates well with Microsoft tools, is powerful and scalable, and provides real-time, interactive insights for better decision-making.

Better Data → Better Decisions
→ **Business Growth**

Different Versions of Power BI & Where They are Used?

What are the versions?

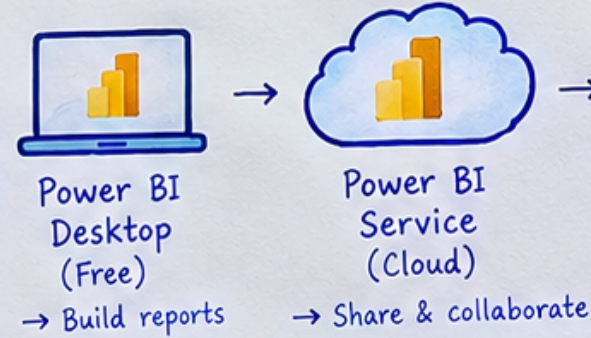
Power BI comes in different versions to meet the needs of different users — from individual analysts to large enterprises. Each version has its own features, target audience and use cases.

Main versions are:

- Power BI Desktop (Free)
- Power BI Service (Cloud)
- Power BI Pro (Paid)
- Power BI Premium (Paid)



Quick Overview



(Licensing Options)

Power BI Pro (Paid)

→ Individual users (share & collaborate)

Power BI Premium (Paid)

→ Large organizations (advanced features & capacity)

1. Power BI Desktop (Free)

Aspect	Details
What it is	A Windows application used to create, design and build reports and dashboards.
Key features	Data modeling, DAX, Power Query, visualizations, report design.
Where it is used	By data analysts, individual users, and developers to prepare and design reports.
Example	An analyst uses Power BI Desktop to connect to an Excel file, clean data using Power Query, create visuals and build a dashboard.
Visual	 Desktop application (install on your computer)

2. Power BI Service (Cloud)

Aspect	Details
What it is	A cloud-based platform to publish, share and collaborate on reports and dashboards.
Key features	Online access, real-time data, workspaces, sharing, scheduling refresh.
Where it is used	By business teams, managers, executives and anyone who needs to view reports from anywhere.
Example	A sales manager accesses the published dashboard in the browser to check daily performance.
Visual	 Web browser (https://app.powerbi.com)

3. Power BI Pro (Paid)

Aspect	Details
What it is	A per-user license that allows access to Power BI Service, sharing and collaboration features.
Key features	Share reports, create workspaces, follow content, set alerts, mobile access.
Where it is used	By individual professionals and small teams who need collaboration and sharing.
Example	A team of 10 analysts buy Pro licenses to share dashboards with their department.
Visual	 User license (for collaboration)

4. Power BI Premium (Paid)

Aspect	Details
What it is	A capacity-based license that provides advanced features, higher limits and enterprise-grade capabilities.
Key features	Larger data capacity, dedicated capacity (Premium), advanced AI features, large datasets, paginated reports.
Where it is used	By large enterprises and organizations with high data volume and complex needs.
Example	A global retail company uses Premium to handle terabytes of sales data, run AI-driven insights and serve thousands of users.
Visual	 Enterprise scale (large capacity & advanced features)

Real World Example – Retail Company

A retail company uses different Power BI versions at various stages:



Result:

- ✓ Better visibility of sales, inventory and customer trends.
- ✓ Faster decision making.
- ✓ Improved operational efficiency and business growth.

In Short...

Each version of Power BI serves a different purpose.

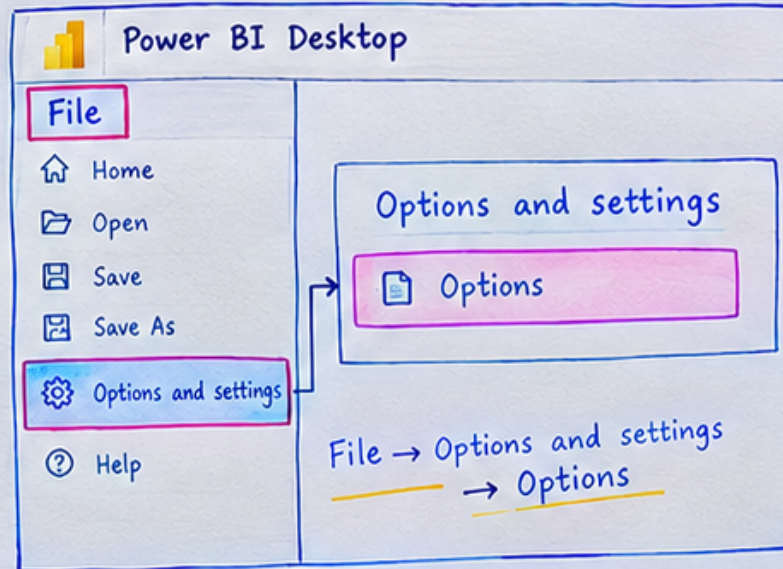
Desktop helps you build, Service helps you share, Pro helps you collaborate, and Premium helps you scale.

Together, they form a complete ecosystem to turn data into valuable insights — for individuals, teams and large enterprises.

Adjusting Settings in Power BI

Steps to Follow

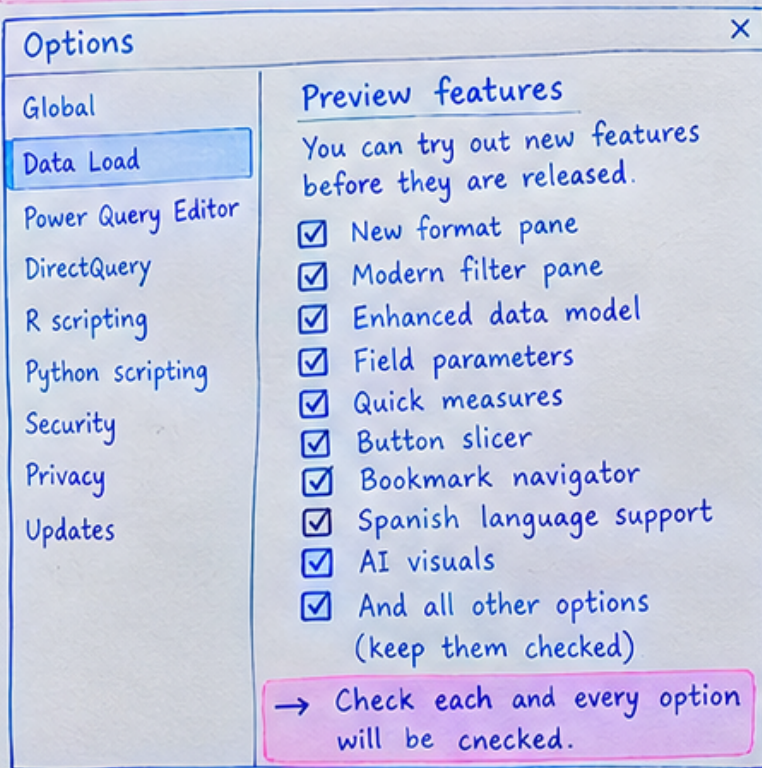
- 1 After installation, open Power BI and go to:
File → Options and settings → Options
- 2 Enable at least one Preview Feature (example: New format pane).
- 3 Make changes in Data Load settings as shown in the video.
- 4 Under the current File, set Regional Settings to India (or your location).



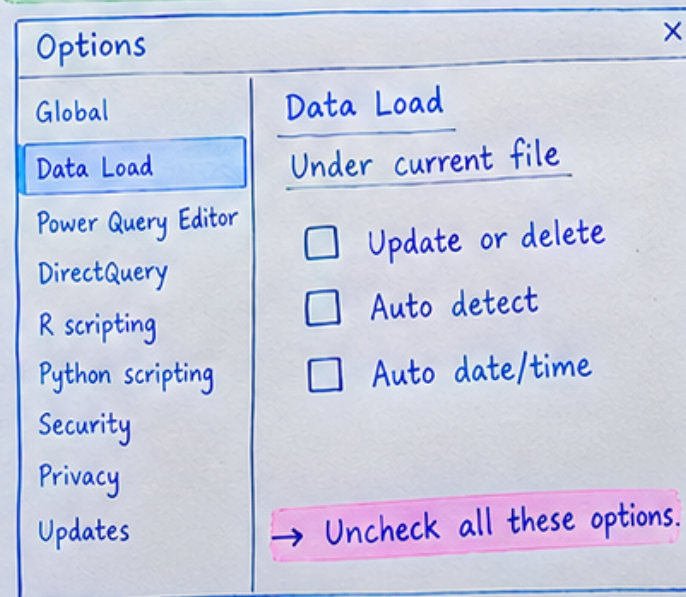
Important

Every time when you open the new Power BI file, we need to change this above settings.

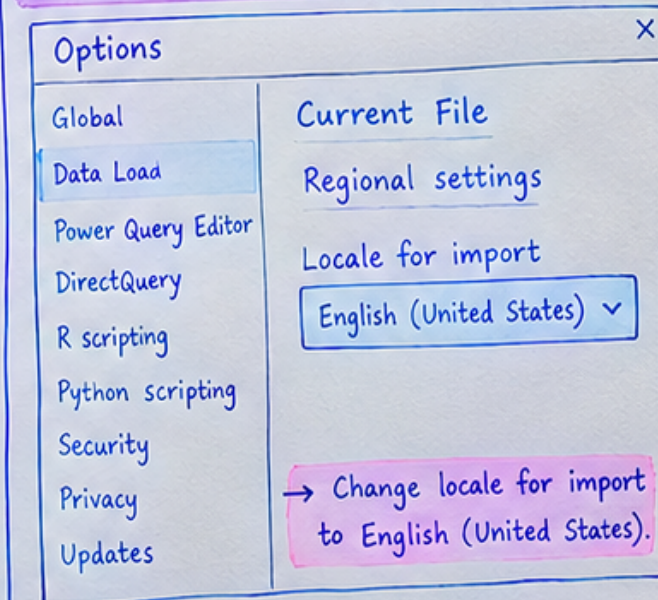
1. Global Preview Feature



2. Current File - Data Load Settings



3. Current File - Regional Setting



Summary of Settings

- 1 Global preview feature → Check all options.
- 2 Current file - Data Load → Uncheck Update or delete, Auto detect, Auto date/time.
- 3 Current file - Regional Setting → English (United States).

★ Why These Settings?

- Enable new features (like New format pane).
 - Avoid unwanted data refresh/update prompts.
 - Prevent automatic date/time detection issues.
 - Ensure correct regional formatting while importing data.
- Helps in smooth data loading and better performance in Power BI.

Report View, Data View and Model View in Power BI

Power-BI has three main views. Each view serves a different purpose and helps you work with your data in a better way.

1. Report View

What is it?

Report View is where you build and design your reports. It is used to create visualizations, arrange them on a canvas, and make your data look meaningful.

Key Features

- Add visuals (charts, tables, cards, etc.)
- Arrange and resize visuals
- Apply filters and slicers
- Format the report (themes, colors, layout)
- Work with multiple pages (tabs)

Visual



Example

- You want to create a sales dashboard with charts, tables and slicers. You switch to Report View, drag visuals from the Visualizations pane and design your report.

2. Data View

What is it?

Data View shows the actual data from your tables, just like a spreadsheet. It helps you check, edit and manage your data directly in Power BI.

Key Features

- View and edit table data
- Check data quality and errors
- Create new columns and measures
- Work with multiple tables
- See relationships between tables

Visual

The screenshot shows the Power BI Data View interface. The top ribbon includes File, Home, and Help. The main canvas displays a table of sales data. A yellow callout box points to the 'Data View icon' (a table icon) in the left-hand Visualizations pane.

OrderID	Customer	Product	Sales	Date
1001	John	Laptop	1200	1/1/2024
1002	Sara	Mobile	850	1/3/2024
1003	Mike	Tablet	600	1/5/2024
1004	Emma	Laptop	1500	1/7/2024
1005	David	Headphones	400	1/9/2024

Example

- You want to check the sales data, fix incorrect values, or create a calculated column. You switch to Data View and work directly with the table data.

3. Model View

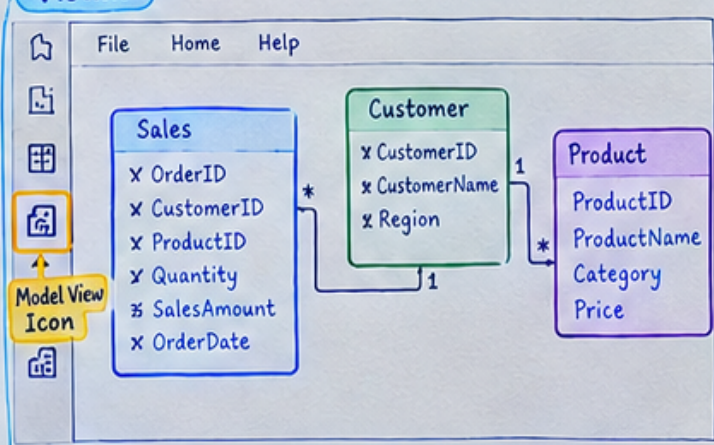
What is it?

Model View is where you manage the data model. It shows how your tables are related and helps you create relationships, calculated columns, measures and more.

Key Features

- Create and manage relationships
- View table structure and data types
- Create measures and calculated columns
- Manage hierarchies
- Handle data modeling and DAX (Data Analysis Expressions)

Visual



Example

- You want to create a relationship between Sales and Customer tables, or define a measure like Total Sales. You switch to Model View and set up the data model.

Quick Comparison

Feature	Report View	Data View	Model View
Main Purpose	Build and design reports	View and edit data	Manage data model
Shows	Visuals on canvas	Table data	Tables, relationships, structure
Used for	Visualization, formatting	Data checking, editing	Relationships, DAX, measures
Icon	(bar chart icon)	(table icon)	(model icon)

In Short...

Report View = Looks good (visuals)
Data View = Checks and edits data
Model View = Connects and manages data

Together, these three views help you build powerful and interactive reports in Power BI.